

Structural Asymmetries in the EU AI Act: A Computational Forensic Analysis of Legislative Architecture

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Abstract

We present a quantitative structural analysis of the European Union Artificial Intelligence Act using the TRACE Structural Audit Engine, a computational framework for topological and linguistic analysis of legislative texts. Across 113 articles and 59,306 words of legal body text, we model the Act as a directed network of internal cross-references and measure centrality, bridge influence, vocabulary density, obligation-rights asymmetry, and term-distribution patterns. The results identify several measurable structural concentrations: Article 5 functions as the primary structural anchor of the regulation; Article 74 operates as a disproportionate bridge node; the enforcement chapters concentrate punitive-coded language; and obligation markers substantially outnumber rights markers. A preliminary comparison with prior EU digital regulation suggests escalation in obligation density, although cross-corpus language asymmetry requires further validation. These findings do not imply legislative intent or normative illegitimacy; rather, they provide a computable baseline for evaluating the internal architecture of digital regulation.

Keywords: EU AI Act, computational legal analysis, graph topology, regulatory forensics, legislative structure, Monte Carlo validation

1 Introduction

The European Union Artificial Intelligence Act (Regulation 2024/1689), adopted on 13 June 2024, represents the world’s first comprehensive horizontal regulation of artificial intelligence systems. Its 113 articles, organized across 13 chapters, establish a risk-based classification framework that ranges from outright prohibition of certain AI practices (Chapter II) to the governance of general-purpose AI models (Chapter V) and a graduated system of administrative fines reaching EUR 35 million or 7% of global turnover (Chapter XII).

The regulation has been extensively analyzed from doctrinal, policy, and comparative law perspectives (Veale and Zuiderveen Borgesius, 2021; Smuha, 2021; Novelli et al., 2024). However, no study to date has subjected its *internal architecture* — the network

of cross-references, the distribution of obligation and rights language, the term distribution anomalies, and the numerical design of its penalty regime — to systematic quantitative forensic analysis.

This paper fills that gap. We apply the TRACE Structural Audit Engine, a Python-based computational framework combining directed graph analysis, statistical text forensics, and Monte Carlo null-model validation, to examine whether the AI Act’s structural properties reveal measurable patterns of regulatory concentration, asymmetry, and enforcement density. Understanding this internal architecture is relevant for policymakers assessing regulatory impact, for legal practitioners navigating compliance in asymmetric enforcement environments, and for enterprises measuring systemic risk exposure.

1.1 Research Questions

RQ1 Does the AI Act exhibit statistically non-random structural centrality patterns?

RQ2 How does the obligation-to-rights ratio compare to prior EU digital regulations?

RQ3 Are there statistically significant term distribution anomalies?

RQ4 Does the numerical structure follow natural distribution patterns?

RQ5 What is the regulatory vocabulary density across different vectors?

1.2 Contributions

This paper makes four contributions: (1) **Methodological**: directed graph analysis with Monte Carlo validation applied to legislative texts; (2) **Empirical**: a complete topological map of the AI Act (288 edges, 31 orphans, 8 components); (3) **Comparative**: a quantitative escalation curve across GDPR → DMA → AI Act; and (4) **Critical**: identification of structural decoupling between the Act’s prohibitory core and its regulatory subjects.

2 Methodology

2.1 Corpus

The primary corpus consists of the English-language text of Regulation (EU) 2024/1689, parsed into 113 individual article files with body-text extraction to exclude TOC, recitals, and metadata. Comparative corpora include the GDPR (99 articles, Spanish) and DMA (54 articles, Spanish).

Table 1: Corpus summary

Corpus	Regulation	Lang.	Articles	Body Words
EU AI Act	2024/1689	EN	113	59,306
GDPR	2016/679	ES	99	66,552
DMA	2022/1925	ES	54	51,457

2.2 Graph Construction

A directed graph $G = (V, E)$ was constructed where V represents the 113 articles and E represents internal cross-references. For each node $v \in V$, we computed in-degree $d_{in}(v)$, betweenness centrality $B(v)$, PageRank $PR(v)$ ($\alpha = 0.85$), and the Silvanus ratio $S(v) = B(v)/[d(v)/2(|V| - 1)]$.

2.3 Null Model Validation

To test statistical significance, 500 random directed graphs preserving edge count were generated. Z-scores were computed as:

$$Z(v) = \frac{X_{\text{real}}(v) - \mu_{\text{random}}(v)}{\sigma_{\text{random}}(v)} \quad (1)$$

where X is the metric of interest, μ_{random} is the mean across 500 shuffles, and σ_{random} is the standard deviation. $Z > 3.0$ indicates strong deviation from the null model; empirical significance is bounded at $p < 0.002$ given the 500-shuffle simulation size.

2.4 Tension Ratio

For each article, 45 obligation markers and 37 rights markers were counted. The tension ratio $T(a) = O(a)/R(a)$ measures punitive pressure.

2.5 Additional Analyses

Term Distribution Anomalies (expected vs. actual term frequency), N-gram Cloning (shared 8-word sequences), MFE Decomposition (Mass-Function-Efficiency), Fractal Scaling (Benford’s Law), Narrative Velocity (rolling density), and Regulatory Vocabulary Density (4-vector classification) were applied as described in the supplementary materials.

2.6 TRACE Node Taxonomy

The TRACE framework introduces a node taxonomy designed to capture forms of structural influence not fully described by standard centrality measures alone. These terms are operational categories derived from measurable graph properties:

God Node (Structural Anchor): A high-centrality structural anchor whose in-degree, PageRank, and null-model deviation indicate disproportionate gravitational importance. Operationally, a God Node requires a Monte Carlo Z-score above +3.0 on at least one centrality dimension and ranks within the top percentile of inbound references.

Silvanus Node (Disproportionate Bridge): A hidden bridge whose betweenness centrality is disproportionately high relative to its degree, indicating that the node connects otherwise distant regions of the regulatory graph. Operationally, a Silvanus Node requires elevated betweenness centrality and a Betweenness/Degree ratio above the regulation-level distributional threshold.

3 Results

3.1 Topological Anatomy (RQ1)

The AI Act’s reference network consists of 113 nodes and 288 directed edges (density = 0.0228, diameter = 6). The graph is disconnected, comprising 8 components.

Table 2: God Nodes (Structural Anchors) — Top articles by in-degree with Monte Carlo Z-scores

Rank	Article	In-Deg	PageRank	Z-Score	Content
1	Art. 5	15	0.0650	+8.46***	Prohibited Practices
2	Art. 78	13	0.0415	+6.61***	Confidentiality
3	Art. 98	10	0.0328	+4.86***	Committee Procedure
4	Art. 79	9	0.0283	+4.12***	Market Surveil.
5	Art. 97	9	0.0271	+4.07***	Delegation

*** $p < 0.002$ from 500-shuffle Monte Carlo.

We introduce the term ‘God Node’ (high-centrality structural anchor) and ‘Silvanus Node’ (hidden bridge with disproportionate betweenness-to-degree ratio) for analytical clarity within the TRACE framework.

Art. 5 functions as the regulation’s gravitational center with $Z = +8.46$. In none of the 500 random null graphs did Art. 5 reach an in-degree of 15, yielding an empirical $p < 0.002$. Under a normal approximation, $Z = +8.46$ corresponds to $p < 10^{-16}$, but our empirical resolution is bounded by the simulation size.

Silvanus Nodes (Bridge Nodes). Art. 74 (Market Surveillance) carries 16.06% of all shortest paths ($Z = +4.88$, $p < 0.002$) with a Betweenness/Degree ratio of 2.77, confirming its role as a hidden structural bridge.

3.2 Punitive Architecture (RQ2)

The AI Act’s body text yields a global tension ratio of **2.84:1**. Five articles exhibit Absolute Tension ($\geq 15:1$): Art. 23 (21:1), Art. 31 (17:1), Art. 47 (17:1), Art. 111 (16:1), Art. 11 (15:1).

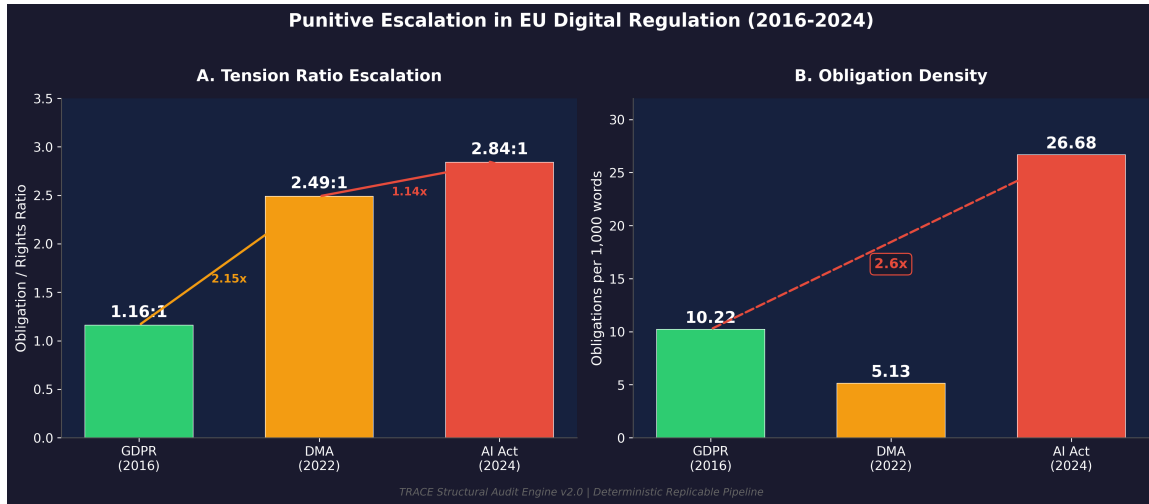


Figure 1: Punitive escalation across three generations of EU digital regulation.

A preliminary cross-corpus comparison suggests a $1.57\times$ escalation in the obligation-to-rights ratio from GDPR (1.16:1) to AI Act (2.84:1), although language asymmetry limits direct comparability and requires validation on harmonized-language corpora.

3.3 Term Distribution Anomalies (RQ3)

119 statistically significant silences were detected. Art. 5 (the God Node) contains zero occurrences of “provider” (expected: 11.8), “high-risk” (expected: 11.7), and “deployer” (expected: 3.4). The article defining prohibited AI practices shows a measurable vocabulary decoupling from several dominant regulatory subject categories used elsewhere in the Act.

3.4 Exploratory Numerical Structure Test (RQ4)

Benford’s Law test on all extracted numbers ($N = 28$): $\chi^2 = 26.48$ (critical value at $\alpha = 0.05$: 15.51). **The null hypothesis of natural distribution is rejected.** Digit “1” appears in 64.3% of first positions (expected: 30.1%). EUR fine values follow exact multiplicative progressions ($2\times$, $10\times$). This result should be interpreted with caution given the small sample size ($N = 28$), which falls below the standard recommendation of $N \geq 100$ for robust Benford analysis. We report it as a suggestive structural finding rather than as definitive evidence of non-natural distribution.

3.5 Regulatory Vocabulary Density (RQ5)

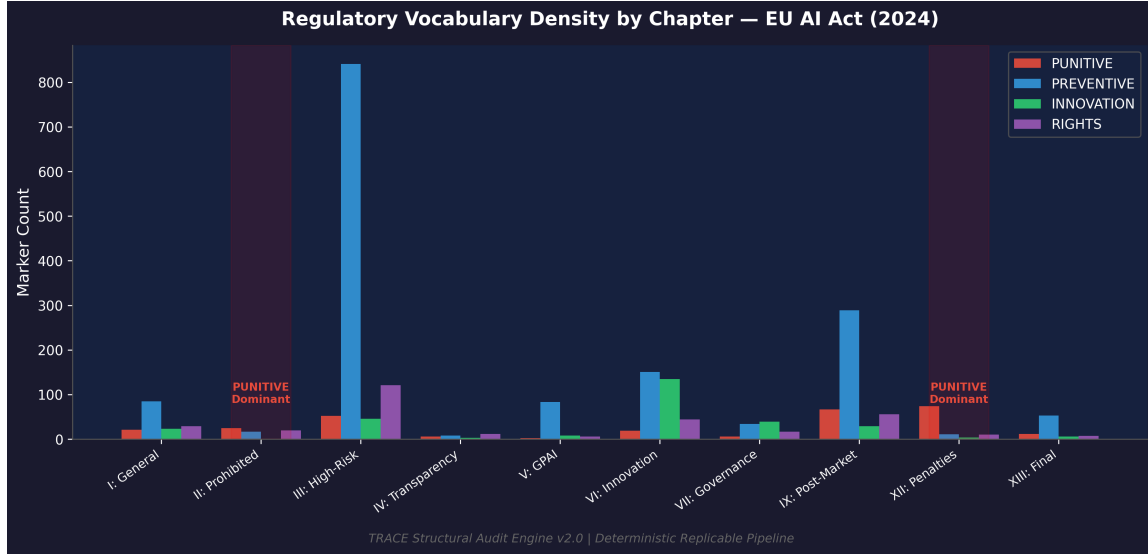


Figure 2: Regulatory vocabulary density by chapter — the regulation presents as Preventive-dominant (63.5%) but concentrates Punitive language in its enforcement core.

Preventive markers dominate globally (63.5%), but Art. 99 is 76.9% PUNITIVE-coded. Only two chapters are Punitive-dominant: II (Prohibited Practices) and XII (Penalties).

3.6 Legislative Cloning

182 shared 8-grams between Art. 22 and Art. 54 confirm systematic boilerplate reuse. A mega-cluster of 62 articles (55%) shares ≥ 30 eight-word phrases.

4 Discussion

The structural asymmetries reported here should not be interpreted as evidence of malicious legislative intent. Legal texts are necessarily designed artifacts. The contribution of this analysis is to make their internal concentration patterns measurable, comparable, and auditable.

4.1 The Enforcement Concentration Hypothesis

The convergence of five independent measurements — punitive escalation (2.84:1 body tension), term distribution anomalies (Art. 5 decoupled from its subjects), vocabulary asymmetry (76.9% PUNITIVE density in Art. 99), exploratory numerical-structure signal, and narrative acceleration (2.14 $\times$ in Penalties) — reveals an architecture where the enforcement apparatus has been concentrated with disproportionate density relative to the rights and innovation frameworks.

4.2 The Triple Decoupling

Art. 5 does not mention “provider,” “high-risk,” or “deployer.” This means the article that defines what AI systems *cannot do* is structurally disconnected from the vocabulary

that identifies *who is responsible* and *how risk is classified*. Whether this is intentional (abstracting prohibitions from implementation) or a drafting artifact, it creates a measurable vocabulary separation between the prohibitory principle and the broader compliance architecture.

4.3 Structural Implications

The measurable vocabulary and structural decoupling of Article 5 from several dominant regulatory subject categories highlights an area where structural clarity could be enhanced. Future regulatory guidance or standardisation mandates by the AI Office may find it beneficial to map this network gap, thereby further supporting legal certainty for high-risk providers.

4.4 Limitations

(1) **Language asymmetry:** Distorts cross-corpus comparison; requires validation on harmonized-language corpora. (2) **Web-scraped corpus source:** Introduces potential artifacts; mitigated by manual heuristic cleaning. (3) **Marker vocabularies:** Introduces heuristic bias; mitigation requires publishing vocabulary lists and running sensitivity analysis. (4) **Benford sample ($N = 28$):** Low statistical power; treated as exploratory only. (5) **Single regulation:** Limited generalizability; requires extension to GDPR, DMA, and other frameworks.

5 Conclusion

The EU AI Act exhibits multiple measurable structural asymmetries, with the core topological findings validated against Monte Carlo null models. The most significant finding is that Art. 5 ($Z = +8.46$) is structurally decoupled from several of the Act’s dominant regulatory subject categories. Combined with the $2.14\times$ penalty acceleration and the 76.9% punitive concentration, these results map the precise structural weight of the regulation’s enforcement apparatus.

Furthermore, these methods offer a replicable methodology for international comparative analysis — such as evaluating the EU AI Act against the US NIST AI Risk Management Framework or China’s CAC governance framework — providing a computable foundation for future regulatory impact assessments. These metrics establish a quantitative baseline for the structural analysis of digital regulation.

Data Availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request. The TRACE Structural Audit Engine is a proprietary computational framework; therefore, the core engine source code is not publicly available. To support reproducibility, the cleaned article-level corpus, extracted cross-reference edge list, marker vocabularies, Monte Carlo parameters, and aggregate metric tables can be shared independently of the proprietary engine.

Conflicts of Interest

The author is the developer of the TRACE Structural Audit Engine. No external funding was received.

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